FAULT CURRENT STUDIO - ENGINEERING REPORT
BROWSER EXPORT - ENGINEERING REVIEW REQUIRED

===== SUMMARY =====
Project: Fault Current Studio advanced study
Method: multisequence
Model status: BOUNDED_MULTI_BUS_SEQUENCE_FAULT
Captured UTC: 2026-10-26T06:04:34.130Z
Warning: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Report data provenance: Entered study inputs, isolated calculated method outputs. No external OEM certification or auto-selected breaker/PPE.

===== STUDY INPUTS =====
Input path: nominalLLV | Entered value: 13800
Input path: voltageFactor | Entered value: 1.1
Input path: frequencyHz | Entered value: 60
Input path: faultType | Entered value: AG
Input path: faultBus | Entered value: F
Input path: buses[0] | Entered value: F
Input path: buses[1] | Entered value: A
Input path: buses[2] | Entered value: B
Input path: branches[0].id | Entered value: line-A
Input path: branches[0].from | Entered value: A
Input path: branches[0].to | Entered value: F
Input path: branches[0].positiveSequence.rOhm | Entered value: 0.06
Input path: branches[0].positiveSequence.xOhm | Entered value: 0.12
Input path: branches[0].negativeSequence.rOhm | Entered value: 0.07
Input path: branches[0].negativeSequence.xOhm | Entered value: 0.14
Input path: branches[0].zeroSequence.rOhm | Entered value: 0.23
Input path: branches[0].zeroSequence.xOhm | Entered value: 0.44
Input path: branches[1].id | Entered value: line-B
Input path: branches[1].from | Entered value: B
Input path: branches[1].to | Entered value: F
Input path: branches[1].positiveSequence.rOhm | Entered value: 0.04
Input path: branches[1].positiveSequence.xOhm | Entered value: 0.13
Input path: branches[1].negativeSequence.rOhm | Entered value: 0.05
Input path: branches[1].negativeSequence.xOhm | Entered value: 0.16
Input path: branches[1].zeroSequence.rOhm | Entered value: 0.22
Input path: branches[1].zeroSequence.xOhm | Entered value: 0.48
Input path: sources[0].id | Entered value: grid-A
Input path: sources[0].bus | Entered value: A
Input path: sources[0].positiveSequence.rOhm | Entered value: 0.08
Input path: sources[0].positiveSequence.xOhm | Entered value: 0.42
Input path: sources[0].negativeSequence.rOhm | Entered value: 0.09
Input path: sources[0].negativeSequence.xOhm | Entered value: 0.46
Input path: sources[0].zeroSequence.rOhm | Entered value: 0.31
Input path: sources[0].zeroSequence.xOhm | Entered value: 1.2
Input path: sources[0].angleDeg | Entered value: 0
Input path: sources[0].emScale | Entered value: 1
Input path: sources[1].id | Entered value: source-B
Input path: sources[1].bus | Entered value: B
Input path: sources[1].positiveSequence.rOhm | Entered value: 0.11
Input path: sources[1].positiveSequence.xOhm | Entered value: 0.6
Input path: sources[1].negativeSequence.rOhm | Entered value: 0.13
Input path: sources[1].negativeSequence.xOhm | Entered value: 0.67
Input path: sources[1].zeroSequence.rOhm | Entered value: 0.33
Input path: sources[1].zeroSequence.xOhm | Entered value: 1.34

Input path: sources[1].angleDeg | Entered value: 0
Input path: sources[1].emScale | Entered value: 1

===== EQUIPMENT DUTY =====
Calculated output path: status | Value: BOUNDED_MULTI_BUS_SEQUENCE_FAULT
Calculated output path: scope | Value: One voltage base, explicitly entered positive/negative/zero-sequence branch and source impedances, independent fixed positive-sequence source EMFs; four metallic fault configurations. No IEC 60909 corrections, transformer vector shifts, time-dependent generator/IBR currents or making/breaking duties.
Calculated output path: faultType | Value: AG
Calculated output path: nominalLLV | Value: 13800
Calculated output path: voltageFactor | Value: 1.1
Calculated output path: frequencyHz | Value: 60
Calculated output path: faultBus | Value: F
Calculated output path: faultImpedanceOhm.rOhm | Value: 0
Calculated output path: faultImpedanceOhm.xOhm | Value: 0
Calculated output path: sequenceTheveninOhm[0].real | Value: 0.2729322097008292
Calculated output path: sequenceTheveninOhm[0].imag | Value: 0.8628023435184697
Calculated output path: sequenceTheveninOhm[0].magnitude | Value: 0.9049419180660937
Calculated output path: sequenceTheveninOhm[0].angleDegrees | Value: 72.44615709955977
Calculated output path: sequenceTheveninOhm[1].real | Value: 0.07332704773129058
Calculated output path: sequenceTheveninOhm[1].imag | Value: 0.3106022392457279
Calculated output path: sequenceTheveninOhm[1].magnitude | Value: 0.31914041886518757
Calculated output path: sequenceTheveninOhm[1].angleDegrees | Value: 76.7168095145296
Calculated output path: sequenceTheveninOhm[2].real | Value: 0.08554316130525336
Calculated output path: sequenceTheveninOhm[2].imag | Value: 0.34845082156908147
Calculated output path: sequenceTheveninOhm[2].magnitude | Value: 0.3587974463374349
Calculated output path: sequenceTheveninOhm[2].angleDegrees | Value: 76.20691177731727
Calculated output path: prefaultPositiveFaultBusV.real | Value: 8764.177086298525
Calculated output path: prefaultPositiveFaultBusV.imag | Value: -1.0694152322308772e-12
Calculated output path: prefaultPositiveFaultBusV.magnitude | Value: 8764.177086298525
Calculated output path: prefaultPositiveFaultBusV.angleDegrees | Value: -6.991298641103816e-15
Calculated output path: sequenceFaultCurrentA[0].real | Value: 1512.2465924481921
Calculated output path: sequenceFaultCurrentA[0].imag | Value: -5329.800273308838
Calculated output path: sequenceFaultCurrentA[0].magnitude | Value: 5540.185981511282
Calculated output path: sequenceFaultCurrentA[0].angleDegrees | Value: -74.15954736119843
Calculated output path: sequenceFaultCurrentA[1].real | Value: 1512.2465924481921
Calculated output path: sequenceFaultCurrentA[1].imag | Value: -5329.800273308838
Calculated output path: sequenceFaultCurrentA[1].magnitude | Value: 5540.185981511282
Calculated output path: sequenceFaultCurrentA[1].angleDegrees | Value: -74.15954736119843
Calculated output path: sequenceFaultCurrentA[2].real | Value: 1512.2465924481921
Calculated output path: sequenceFaultCurrentA[2].imag | Value: -5329.800273308838
Calculated output path: sequenceFaultCurrentA[2].magnitude | Value: 5540.185981511282
Calculated output path: sequenceFaultCurrentA[2].angleDegrees | Value: -74.15954736119843
Calculated output path: phaseFaultCurrentA[0].real | Value: 4536.739777344576
Calculated output path: phaseFaultCurrentA[0].imag | Value: -15989.400819926514
Calculated output path: phaseFaultCurrentA[0].magnitude | Value: 16620.557944533848
Calculated output path: phaseFaultCurrentA[0].angleDegrees | Value: -74.15954736119843
Calculated output path: phaseFaultCurrentA[1].real | Value: 4.547473508864641e-13
Calculated output path: phaseFaultCurrentA[1].imag | Value: 0
Calculated output path: phaseFaultCurrentA[1].magnitude | Value: 4.547473508864641e-13
Calculated output path: phaseFaultCurrentA[1].angleDegrees | Value: 0
Calculated output path: phaseFaultCurrentA[2].real | Value: 0
Calculated output path: phaseFaultCurrentA[2].imag | Value: 0
Calculated output path: phaseFaultCurrentA[2].magnitude | Value: 0
Calculated output path: phaseFaultCurrentA[2].angleDegrees | Value: 0
Calculated output path: groundResidualA | Value: 16620.557944533844
Calculated output path: maximumPhaseRmsA | Value: 16620.557944533848
Calculated output path: sourceContributions[0].id | Value: grid-A

Calculated output path: sourceContributions[0].bus | Value: A
Calculated output path: sourceContributions[0].phaseCurrentsA[0].real | Value: 2691.562397669685
Calculated output path: sourceContributions[0].phaseCurrentsA[0].imag | Value: -8875.047550729882
Calculated output path: sourceContributions[0].phaseCurrentsA[0].magnitude | Value:
9274.210325858794
Calculated output path: sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value:
-73.12886548949481
Calculated output path: sourceContributions[0].phaseCurrentsA[1].real | Value: -80.07152598509356
Calculated output path: sourceContributions[0].phaseCurrentsA[1].imag | Value: 264.44816588912727
Calculated output path: sourceContributions[0].phaseCurrentsA[1].magnitude | Value:
276.304689999473
Calculated output path: sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value:
106.8456418478821
Calculated output path: sourceContributions[0].phaseCurrentsA[2].real | Value: -135.7143107704046
Calculated output path: sourceContributions[0].phaseCurrentsA[2].imag | Value: 258.7728188264974
Calculated output path: sourceContributions[0].phaseCurrentsA[2].magnitude | Value:
292.20155015211196
Calculated output path: sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value:
117.67492081629787
Calculated output path: sourceContributions[0].sequenceCurrentsA[0].real | Value: 825.2588536380623
Calculated output path: sourceContributions[0].sequenceCurrentsA[0].imag | Value:
-2783.942188671419
Calculated output path: sourceContributions[0].sequenceCurrentsA[0].magnitude | Value:
2903.684949400093
Calculated output path: sourceContributions[0].sequenceCurrentsA[0].angleDegrees | Value:
-73.48831515028421
Calculated output path: sourceContributions[0].sequenceCurrentsA[1].real | Value: 931.5134404386342
Calculated output path: sourceContributions[0].sequenceCurrentsA[1].imag | Value:
-3029.4899926421012
Calculated output path: sourceContributions[0].sequenceCurrentsA[1].magnitude | Value:
3169.467921471435
Calculated output path: sourceContributions[0].sequenceCurrentsA[1].angleDegrees | Value:
-72.90828276612866
Calculated output path: sourceContributions[0].sequenceCurrentsA[2].real | Value: 934.7901035929883
Calculated output path: sourceContributions[0].sequenceCurrentsA[2].imag | Value:
-3061.6153694163613
Calculated output path: sourceContributions[0].sequenceCurrentsA[2].magnitude | Value:
3201.1437343583734
Calculated output path: sourceContributions[0].sequenceCurrentsA[2].angleDegrees | Value:
-73.02121840574361
Calculated output path: sourceContributions[1].id | Value: source-B
Calculated output path: sourceContributions[1].bus | Value: B
Calculated output path: sourceContributions[1].phaseCurrentsA[0].real | Value: 1845.177379674894
Calculated output path: sourceContributions[1].phaseCurrentsA[0].imag | Value: -7114.353269196625
Calculated output path: sourceContributions[1].phaseCurrentsA[0].magnitude | Value:
7349.741628206573
Calculated output path: sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value:
-75.4601462566015
Calculated output path: sourceContributions[1].phaseCurrentsA[1].real | Value: 80.07152598510766
Calculated output path: sourceContributions[1].phaseCurrentsA[1].imag | Value: -264.44816588913363
Calculated output path: sourceContributions[1].phaseCurrentsA[1].magnitude | Value:
276.3046899994832
Calculated output path: sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value:
-73.15435815211549
Calculated output path: sourceContributions[1].phaseCurrentsA[2].real | Value: 135.7143107703905
Calculated output path: sourceContributions[1].phaseCurrentsA[2].imag | Value: -258.77281882651005
Calculated output path: sourceContributions[1].phaseCurrentsA[2].magnitude | Value:
292.2015501521166
Calculated output path: sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value:

-62.32507918370572
Calculated output path: sourceContributions[1].sequenceCurrentsA[0].real | Value: 686.9877388101307
Calculated output path: sourceContributions[1].sequenceCurrentsA[0].imag | Value:
-2545.8580846374225
Calculated output path: sourceContributions[1].sequenceCurrentsA[0].magnitude | Value:
2636.9197068528993
Calculated output path: sourceContributions[1].sequenceCurrentsA[0].angleDegrees | Value:
-74.8986886937207
Calculated output path: sourceContributions[1].sequenceCurrentsA[1].real | Value: 580.7331520095568
Calculated output path: sourceContributions[1].sequenceCurrentsA[1].imag | Value:
-2300.3102806667225
Calculated output path: sourceContributions[1].sequenceCurrentsA[1].magnitude | Value:
2372.4835892338583
Calculated output path: sourceContributions[1].sequenceCurrentsA[1].angleDegrees | Value:
-75.83125142278031
Calculated output path: sourceContributions[1].sequenceCurrentsA[2].real | Value: 577.4564888552064
Calculated output path: sourceContributions[1].sequenceCurrentsA[2].imag | Value:
-2268.184903892479
Calculated output path: sourceContributions[1].sequenceCurrentsA[2].magnitude | Value:
2340.5381335852485
Calculated output path: sourceContributions[1].sequenceCurrentsA[2].angleDegrees | Value:
-75.71652407446481
Calculated output path: branchCurrents[0].id | Value: line-A
Calculated output path: branchCurrents[0].from | Value: A
Calculated output path: branchCurrents[0].to | Value: F
Calculated output path: branchCurrents[0].phaseCurrentsA[0].real | Value: 2691.5623976696897
Calculated output path: branchCurrents[0].phaseCurrentsA[0].imag | Value: -8875.047550729887
Calculated output path: branchCurrents[0].phaseCurrentsA[0].magnitude | Value: 9274.2103258588
Calculated output path: branchCurrents[0].phaseCurrentsA[0].angleDegrees | Value:
-73.12886548949479
Calculated output path: branchCurrents[0].phaseCurrentsA[1].real | Value: -80.07152598510356
Calculated output path: branchCurrents[0].phaseCurrentsA[1].imag | Value: 264.44816588912636
Calculated output path: branchCurrents[0].phaseCurrentsA[1].magnitude | Value: 276.30468999947504
Calculated output path: branchCurrents[0].phaseCurrentsA[1].angleDegrees | Value:
106.84564184788417
Calculated output path: branchCurrents[0].phaseCurrentsA[2].real | Value: -135.71431077039915
Calculated output path: branchCurrents[0].phaseCurrentsA[2].imag | Value: 258.772818826507
Calculated output path: branchCurrents[0].phaseCurrentsA[2].magnitude | Value: 292.20155015211793
Calculated output path: branchCurrents[0].phaseCurrentsA[2].angleDegrees | Value:
117.67492081629607
Calculated output path: branchCurrents[0].sequenceCurrentsA[0].real | Value: 825.2588536380622
Calculated output path: branchCurrents[0].sequenceCurrentsA[0].imag | Value: -2783.942188671418
Calculated output path: branchCurrents[0].sequenceCurrentsA[0].magnitude | Value:
2903.6849494000917
Calculated output path: branchCurrents[0].sequenceCurrentsA[0].angleDegrees | Value:
-73.48831515028421
Calculated output path: branchCurrents[0].sequenceCurrentsA[1].real | Value: 931.5134404386398
Calculated output path: branchCurrents[0].sequenceCurrentsA[1].imag | Value: -3029.4899926421094
Calculated output path: branchCurrents[0].sequenceCurrentsA[1].magnitude | Value:
3169.4679214714442
Calculated output path: branchCurrents[0].sequenceCurrentsA[1].angleDegrees | Value:
-72.90828276612862
Calculated output path: branchCurrents[0].sequenceCurrentsA[2].real | Value: 934.7901035929874
Calculated output path: branchCurrents[0].sequenceCurrentsA[2].imag | Value: -3061.61536941636
Calculated output path: branchCurrents[0].sequenceCurrentsA[2].magnitude | Value: 3201.143734358372
Calculated output path: branchCurrents[0].sequenceCurrentsA[2].angleDegrees | Value:
-73.02121840574361
Calculated output path: branchCurrents[1].id | Value: line-B
Calculated output path: branchCurrents[1].from | Value: B

Calculated output path: branchCurrents[1].to | Value: F
Calculated output path: branchCurrents[1].phaseCurrentsA[0].real | Value: 1845.1773796749053
Calculated output path: branchCurrents[1].phaseCurrentsA[0].imag | Value: -7114.353269196645
Calculated output path: branchCurrents[1].phaseCurrentsA[0].magnitude | Value: 7349.741628206596
Calculated output path: branchCurrents[1].phaseCurrentsA[0].angleDegrees | Value:
-75.46014625660146
Calculated output path: branchCurrents[1].phaseCurrentsA[1].real | Value: 80.07152598508105
Calculated output path: branchCurrents[1].phaseCurrentsA[1].imag | Value: -264.44816588913136
Calculated output path: branchCurrents[1].phaseCurrentsA[1].magnitude | Value: 276.3046899994733
Calculated output path: branchCurrents[1].phaseCurrentsA[1].angleDegrees | Value:
-73.15435815212062
Calculated output path: branchCurrents[1].phaseCurrentsA[2].real | Value: 135.71431077040643
Calculated output path: branchCurrents[1].phaseCurrentsA[2].imag | Value: -258.77281882648913
Calculated output path: branchCurrents[1].phaseCurrentsA[2].magnitude | Value: 292.2015501521055
Calculated output path: branchCurrents[1].phaseCurrentsA[2].angleDegrees | Value:
-62.32507918370105
Calculated output path: branchCurrents[1].sequenceCurrentsA[0].real | Value: 686.987738810131
Calculated output path: branchCurrents[1].sequenceCurrentsA[0].imag | Value: -2545.8580846374216
Calculated output path: branchCurrents[1].sequenceCurrentsA[0].magnitude | Value:
2636.9197068528983
Calculated output path: branchCurrents[1].sequenceCurrentsA[0].angleDegrees | Value:
-74.89868869372069
Calculated output path: branchCurrents[1].sequenceCurrentsA[1].real | Value: 580.7331520095677
Calculated output path: branchCurrents[1].sequenceCurrentsA[1].imag | Value: -2300.3102806667453
Calculated output path: branchCurrents[1].sequenceCurrentsA[1].magnitude | Value: 2372.483589233883
Calculated output path: branchCurrents[1].sequenceCurrentsA[1].angleDegrees | Value:
-75.8312514227802
Calculated output path: branchCurrents[1].sequenceCurrentsA[2].real | Value: 577.4564888552067
Calculated output path: branchCurrents[1].sequenceCurrentsA[2].imag | Value: -2268.1849038924775
Calculated output path: branchCurrents[1].sequenceCurrentsA[2].magnitude | Value:
2340.5381335852467
Calculated output path: branchCurrents[1].sequenceCurrentsA[2].angleDegrees | Value:
-75.71652407446481
Calculated output path: busVoltages[0].bus | Value: F
Calculated output path: busVoltages[0].phaseVoltagesV[0].real | Value: 1.5916157281026244e-12
Calculated output path: busVoltages[0].phaseVoltagesV[0].imag | Value: 4.689582056016661e-13
Calculated output path: busVoltages[0].phaseVoltagesV[0].magnitude | Value: 1.6592656280851425e-12
Calculated output path: busVoltages[0].phaseVoltagesV[0].angleDegrees | Value: 16.417232830434777
Calculated output path: busVoltages[0].phaseVoltagesV[1].real | Value: -7523.775721922144
Calculated output path: busVoltages[0].phaseVoltagesV[1].imag | Value: -7555.841674042445
Calculated output path: busVoltages[0].phaseVoltagesV[1].magnitude | Value: 10662.923825901677
Calculated output path: busVoltages[0].phaseVoltagesV[1].angleDegrees | Value: -134.87816395435416
Calculated output path: busVoltages[0].phaseVoltagesV[2].real | Value: -7510.139189234899
Calculated output path: busVoltages[0].phaseVoltagesV[2].imag | Value: 8005.554459790879
Calculated output path: busVoltages[0].phaseVoltagesV[2].magnitude | Value: 10976.843483003639
Calculated output path: busVoltages[0].phaseVoltagesV[2].angleDegrees | Value: 133.17117015805255
Calculated output path: busVoltages[0].sequenceVoltagesV[0].real | Value: -5011.3049703856805
Calculated output path: busVoltages[0].sequenceVoltagesV[0].imag | Value: 149.90426191614415
Calculated output path: busVoltages[0].sequenceVoltagesV[0].magnitude | Value: 5013.546528551706
Calculated output path: busVoltages[0].sequenceVoltagesV[0].angleDegrees | Value:
178.28660973836136
Calculated output path: busVoltages[0].sequenceVoltagesV[1].real | Value: 6997.840608610377
Calculated output path: busVoltages[0].sequenceVoltagesV[1].imag | Value: -78.8886588669685
Calculated output path: busVoltages[0].sequenceVoltagesV[1].magnitude | Value: 6998.285261691915
Calculated output path: busVoltages[0].sequenceVoltagesV[1].angleDegrees | Value:
-0.6458843512941284
Calculated output path: busVoltages[0].sequenceVoltagesV[2].real | Value: -1986.5356382246948
Calculated output path: busVoltages[0].sequenceVoltagesV[2].imag | Value: -71.01560304917518
Calculated output path: busVoltages[0].sequenceVoltagesV[2].magnitude | Value: 1987.804582400703

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Calculated output path: busVoltages[0].sequenceVoltagesV[2].angleDegrees | Value:
-177.95263558388115
Calculated output path: busVoltages[1].bus | Value: A
Calculated output path: busVoltages[1].phaseVoltagesV[0].real | Value: 2328.235163865351
Calculated output path: busVoltages[1].phaseVoltagesV[0].imag | Value: -430.62305585569516
Calculated output path: busVoltages[1].phaseVoltagesV[0].magnitude | Value: 2367.723631358528
Calculated output path: busVoltages[1].phaseVoltagesV[0].angleDegrees | Value: -10.478827958056545
Calculated output path: busVoltages[1].phaseVoltagesV[1].real | Value: -6554.12506477979
Calculated output path: busVoltages[1].phaseVoltagesV[1].imag | Value: -7691.686276806313
Calculated output path: busVoltages[1].phaseVoltagesV[1].magnitude | Value: 10105.37446835026
Calculated output path: busVoltages[1].phaseVoltagesV[1].angleDegrees | Value: -130.43448596658666
Calculated output path: busVoltages[1].phaseVoltagesV[2].real | Value: -6563.792712186067
Calculated output path: busVoltages[1].phaseVoltagesV[2].imag | Value: 7740.443695029404
Calculated output path: busVoltages[1].phaseVoltagesV[2].magnitude | Value: 10148.785314729412
Calculated output path: busVoltages[1].phaseVoltagesV[2].angleDegrees | Value: 130.29749874525513
Calculated output path: busVoltages[1].sequenceVoltagesV[0].real | Value: -3596.5608710335023
Calculated output path: busVoltages[1].sequenceVoltagesV[0].imag | Value: -127.28854587753472
Calculated output path: busVoltages[1].sequenceVoltagesV[0].magnitude | Value: 3598.8126476604593
Calculated output path: busVoltages[1].sequenceVoltagesV[0].angleDegrees | Value:
-177.97304871060717
Calculated output path: busVoltages[1].sequenceVoltagesV[1].real | Value: 7417.270214153748
Calculated output path: busVoltages[1].sequenceVoltagesV[1].imag | Value: -148.87644557285827
Calculated output path: busVoltages[1].sequenceVoltagesV[1].magnitude | Value: 7418.764157581692
Calculated output path: busVoltages[1].sequenceVoltagesV[1].angleDegrees | Value: -1.14986329757395
Calculated output path: busVoltages[1].sequenceVoltagesV[2].real | Value: -1492.4741792548953
Calculated output path: busVoltages[1].sequenceVoltagesV[2].imag | Value: -154.45806440530214
Calculated output path: busVoltages[1].sequenceVoltagesV[2].magnitude | Value: 1500.4454236667211
Calculated output path: busVoltages[1].sequenceVoltagesV[2].angleDegrees | Value:
-174.09142098368295
Calculated output path: busVoltages[2].bus | Value: B
Calculated output path: busVoltages[2].phaseVoltagesV[0].real | Value: 2087.201254796809
Calculated output path: busVoltages[2].phaseVoltagesV[0].imag | Value: -267.8679724345863
Calculated output path: busVoltages[2].phaseVoltagesV[0].magnitude | Value: 2104.3199207063526
Calculated output path: busVoltages[2].phaseVoltagesV[0].angleDegrees | Value: -7.313269217975994
Calculated output path: busVoltages[2].phaseVoltagesV[1].real | Value: -6503.7562349062455
Calculated output path: busVoltages[2].phaseVoltagesV[1].imag | Value: -7707.209879457681
Calculated output path: busVoltages[2].phaseVoltagesV[1].magnitude | Value: 10084.638282511274
Calculated output path: busVoltages[2].phaseVoltagesV[1].angleDegrees | Value: -130.1594643542308
Calculated output path: busVoltages[2].phaseVoltagesV[2].real | Value: -6497.912381555031
Calculated output path: busVoltages[2].phaseVoltagesV[2].imag | Value: 7733.78664566659
Calculated output path: busVoltages[2].phaseVoltagesV[2].magnitude | Value: 10101.20394799833
Calculated output path: busVoltages[2].phaseVoltagesV[2].angleDegrees | Value: 130.03691132992225
Calculated output path: busVoltages[2].sequenceVoltagesV[0].real | Value: -3638.1557872214894
Calculated output path: busVoltages[2].sequenceVoltagesV[0].imag | Value: -80.43040207522574
Calculated output path: busVoltages[2].sequenceVoltagesV[0].magnitude | Value: 3639.044734771915
Calculated output path: busVoltages[2].sequenceVoltagesV[0].angleDegrees | Value:
-178.73354185037942
Calculated output path: busVoltages[2].sequenceVoltagesV[1].real | Value: 7320.1102711774365
Calculated output path: busVoltages[2].sequenceVoltagesV[1].imag | Value: -95.40576033239454
Calculated output path: busVoltages[2].sequenceVoltagesV[1].magnitude | Value: 7320.731974420454
Calculated output path: busVoltages[2].sequenceVoltagesV[1].angleDegrees | Value:
-0.7467152428685605
Calculated output path: busVoltages[2].sequenceVoltagesV[2].real | Value: -1594.753229159138
Calculated output path: busVoltages[2].sequenceVoltagesV[2].imag | Value: -92.03181002696601
Calculated output path: busVoltages[2].sequenceVoltagesV[2].magnitude | Value: 1597.4065593862879
Calculated output path: busVoltages[2].sequenceVoltagesV[2].angleDegrees | Value:
-176.6971740846384
Calculated output path: kclRelativeResidual | Value: 2.634614910300192e-15
Calculated output path: nodalRelativeResidual | Value: 1.1443916996305582e-15
```

Calculated output path: limitations[0] | Value: Explicit positive/negative/zero-sequence equivalent impedances; no transformer winding shift or zero-sequence grounding topology inference
Calculated output path: limitations[1] | Value: Multiple documented fixed positive-sequence Thevenin EMFs, one voltage level only
Calculated output path: limitations[2] | Value: No motor/IBR dynamic or generator subtransient decrement; no short-circuit peak, making/breaking duty
Calculated output path: limitations[3] | Value: No IEC 60909 voltage correction by equipment type, maximum/minimum operating envelope or breaker rating
Calculated output path: limitations[4] | Value: No harmonic, nonlinear electric arc, fault resistance estimation or conductor thermal qualification

===== ASSUMPTIONS & QA =====
Evidence / limitation: Release status | Value: BOUNDED METHOD, NOT A STANDARDS/DEVICE APPROVAL
Evidence / limitation: Warning | Value: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Evidence / limitation: Study method | Value: multisequence
Evidence / limitation: Timestamp | Value: 2026-10-26T06:04:34.130Z
Evidence / limitation: outputs.scope | Value: One voltage base, explicitly entered positive/negative/zero-sequence branch and source impedances, independent fixed positive-sequence source EMFs; four metallic fault configurations. No IEC 60909 corrections, transformer vector shifts, time-dependent generator/IBR currents or making/breaking duties.
Evidence / limitation: outputs.limitations[0] | Value: Explicit positive/negative/zero-sequence equivalent impedances; no transformer winding shift or zero-sequence grounding topology inference
Evidence / limitation: outputs.limitations[1] | Value: Multiple documented fixed positive-sequence Thevenin EMFs, one voltage level only
Evidence / limitation: outputs.limitations[2] | Value: No motor/IBR dynamic or generator subtransient decrement; no short-circuit peak, making/breaking duty
Evidence / limitation: outputs.limitations[3] | Value: No IEC 60909 voltage correction by equipment type, maximum/minimum operating envelope or breaker rating
Evidence / limitation: outputs.limitations[4] | Value: No harmonic, nonlinear electric arc, fault resistance estimation or conductor thermal qualification